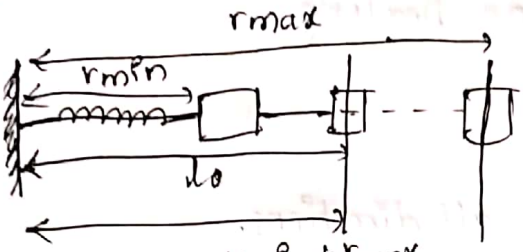
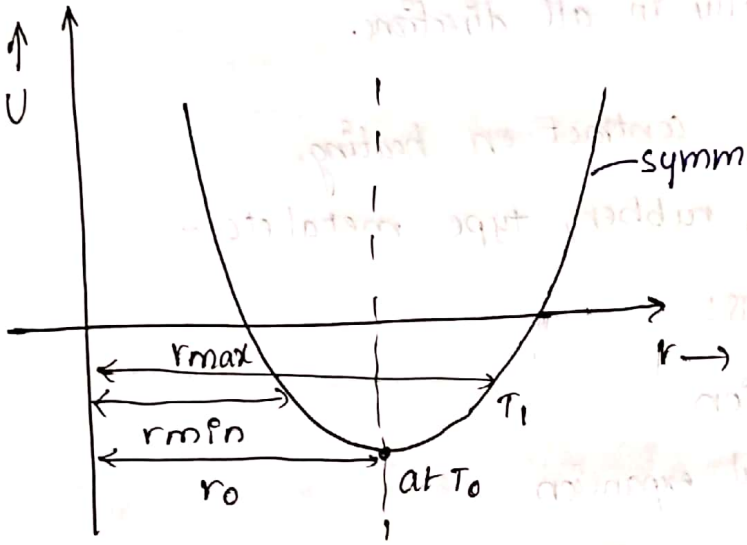
Thermal Expansion of solids.

① Harmonic Vibrations :



$$r_{avg} = \frac{r_{min} + r_{max}}{2}$$

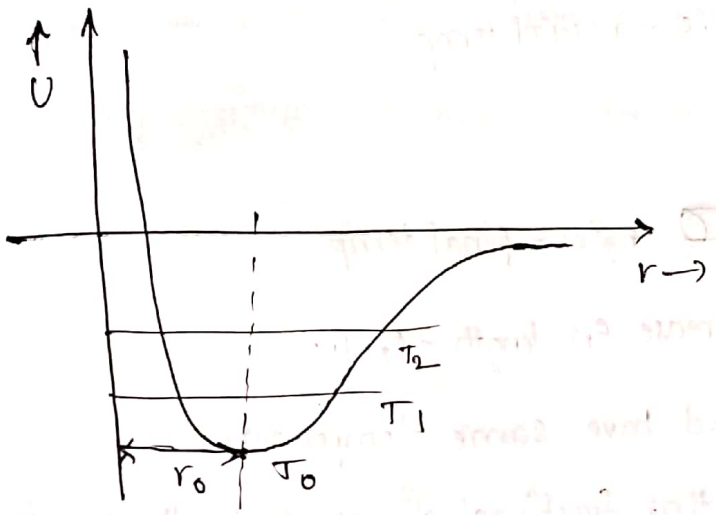


* $r_{avg} = \frac{r_{min} + r_{max}}{2}$

└ same at all temperatures.

* so these type of solids do not expand on heating.

② Anharmonic Vibrations :



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* As temp increases,

r_{avg} also increases.

* These solids expand on heating.

* Isotropic solids:

expands equally in all directions.

* Anisotropic solids:

expands unequally in all directions.

* few substances contract on heating.

Ex: Cast iron, rubber, type metal etc..

Types of expansions:

① linear expansion

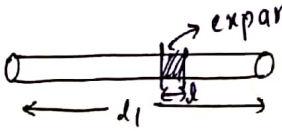
② Areal / superficial expansion

③ volume / cubical expansion

1) Linear expansion:

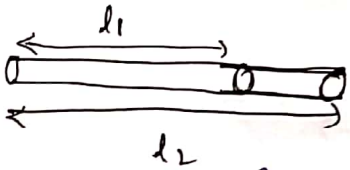
Increase in length of the solids on heating is called linear expansion.

Ex: rods, wires etc..

* 

$t_1°c$ = initial temp

when heated



$t_2°c$ = final temp

increase in length = $l_2 - l_1$.

* every element in rod have same temperature.

* It is observed that fractional change in length is directly

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proportional to rise in temp.

$$\frac{dl}{l} \propto dt$$

$$\frac{dl}{l} = \alpha\, dt$$

α = coeff of linear expansion

① α = const ② $\alpha = f(x)$ ③ $\alpha = f(t)$ (↓ temp.)

$$\int_{l_1}^{l_2} \frac{dl}{l} = \int_{t_1}^{t_2} \alpha \cdot dt \quad (\alpha = \text{const})$$

$$\ln\left(\frac{l_2}{l_1}\right) = \alpha(t_2 - t_1)$$

$$\frac{l_2}{l_1} = e^{\alpha(t_2-t_1)}$$

$$\frac{l_2}{l_1} = \left[1 + \alpha(t_2-t_1) + \alpha^2 \frac{(t_2-t_1)^2}{2!} \cdots\right]$$

(α is very small)

$$\frac{l_2}{l_1} = 1 + \alpha(t_2-t_1)$$

$$\boxed{l_2 = l_1\left[1 + \alpha(t_2-t_1)\right]}$$

⇒ linear expansion,

$$l_2 - l_1 = \boxed{\Delta l = l_1 \alpha (t_2-t_1)}$$

* fractional change in length,

$$\boxed{\frac{\Delta l}{l_1} = \alpha(t_2-t_1)} = \frac{l_2-l_1}{l_1}$$

* percentage of change in length,

$$\frac{\Delta l}{l} \times 100 = \alpha(t_2-t_1) \times 100\%$$

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Again, $\alpha = \frac{l_2 - l_1}{l_1(t_2 - t_1)}$

* α is defined as fractional change in length per change in temp.

* units: $/^\circ C$ (or) $/K$

* α - depend on: 1) nature of material
2) on unit of temp.

α-values for some materials:

lead $\longrightarrow 29 \times 10^{-6}/^\circ C$

aluminium $\longrightarrow 24 \times 10^{-6}/^\circ C$

Brass $\longrightarrow 19 \times 10^{-6}/^\circ C$

Copper $\longrightarrow 17 \times 10^{-6}/^\circ C$

Steel | Iron $\longrightarrow 12 \times 10^{-6}/^\circ C$

glass $\longrightarrow 9 \times 10^{-6}/^\circ C$

Invar steel $\longrightarrow 0.9 \times 10^{-6}/^\circ C$

* α is more then that substance expands more on heating & contracts more on cooling.

* A temp diff of $100^\circ C = 180^\circ F$

$1^\circ C = 1.8^\circ F$

ex: for lead: $\alpha = 29 \times 10^{-6}/^\circ C$

in $^\circ F \Rightarrow \alpha = \frac{29 \times 10^{-6}}{1.8}/^\circ F$

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* α is defined for ⟶ length $\Rightarrow l_2 = l_1(1+\alpha(t_2-t_1))$

breadth $\Rightarrow b_2 = b_1(1+\alpha(t_2-t_1))$

thickness $\Rightarrow h_2 = h_1(1+\alpha(t_2-t_1))$

Radius $\Rightarrow R_2 = R_1(1+\alpha(t_2-t_1))$

radius of gyration $\Rightarrow k_2 = k_1(1+\alpha(t_2-t_1))$

Effect of temperature on rotating body:

① Radius of gyration :-

$$k_2 = k_1(1+\alpha(t_2-t_1))$$

② Moment of inertia :-

$$I = MK^2$$

M = mass of body which remains same on heating.

$\Rightarrow I_1 = MK_1^2$ (before heating)

& $I_2 = MK_2^2$ (after heating)

$$I_2 = MK_1^2(1+\alpha(t_2-t_1))^2$$

$$\boxed{I_2 = I_1(1+2\alpha(t_2-t_1))}$$

$(1+x)^n \approx 1+nx$

as temp increases M.I of body also increases

* $\frac{I_2-I_1}{I_1} = \frac{\Delta I}{I} = 2\alpha(t_2-t_1)$

* % of increase in moment of inertia,

$$\frac{\Delta I}{I} \times 100 = 2\alpha(t_2-t_1) \times 100\%$$

③ Angular velocity :

we have, $L = I\omega$

* L is not changed on heating.

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$$\Delta L = I \Delta\omega + \omega \Delta I$$

$$\frac{\Delta L}{L} = \frac{\Delta\omega}{\omega} + \frac{\Delta I}{I}$$

$$\frac{\Delta\omega}{\omega} + \frac{\Delta I}{I} = 0$$

$$\frac{\Delta\omega}{\omega} = -\frac{\Delta I}{I}$$

$$\boxed{\frac{\Delta\omega}{\omega} = -2\alpha(t_2 - t_1)}$$

* % of change in ω,

$$\frac{\Delta\omega}{\omega} \times 100 = -2\alpha(t_2 - t_1) \times 100$$

* $$\frac{\omega_2 - \omega_1}{\omega_1} = -2\alpha(t_2 - t_1)$$

$$\boxed{\omega_2 = \omega_1 (1 - 2\alpha(t_2 - t_1))}$$

i.e, angular velocity decreases on heating.

④ Rotational Kinetic Energy:

we have,

$$K = \frac{L^2}{2I}$$

$$K = \frac{L^2}{2}(I^{-1})$$

$$\Delta K = \frac{L^2}{2}\left(-\frac{1}{I^2}\right) \cdot \Delta I$$

$$\Delta K = -\frac{L^2}{2I^2}\Delta I$$

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$$\frac{\Delta K}{K} = \frac{\frac{-V^2}{2I^2}\Delta I}{\frac{V^2}{2I}}$$

$$\frac{\Delta K}{K} = -\frac{\Delta I}{I}$$

$$\boxed{\frac{\Delta K}{K} = -2\alpha(t_2 - t_1)}$$

* % of change in K,

$$\frac{\Delta K}{K} \times 100 = -2\alpha(t_2 - t_1) \times 100$$

$$\frac{K_2 - K_1}{K_1} = -2\alpha(t_2 - t_1)$$

$$\boxed{K_2 = K_1(1 - 2\alpha(t_2 - t_1))}$$

i.e, rotational K.E decreases on heating.

⑤ Linear Velocity (or) tangential velocity:

$$V = R\omega$$

$$\Delta V = R\Delta\omega + \omega\Delta R$$

$$\frac{\Delta V}{V} = \frac{\Delta\omega}{\omega} + \frac{\Delta R}{R}$$

$$\frac{\Delta V}{V} = -2\alpha(t_2 - t_1) + \alpha(t_2 - t_1)$$

$$\boxed{\frac{\Delta V}{V} = -\alpha(t_2 - t_1)}$$

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* $\boxed{\frac{\Delta V}{V} \times 100 = -\alpha(t_2 - t_1) \times 100}$

* $\frac{V_2 - V_1}{V_1} = -\alpha(t_2 - t_1)$

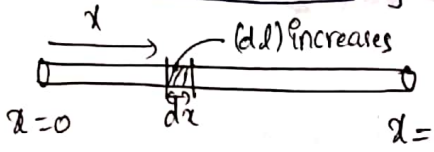
$$\boxed{V_2 = V_1(1 - \alpha(t_2 - t_1))}$$

i.e, linear velocity decreases on heating.

Variable α :-

① $\alpha = f(x)$

for ex, $\boxed{\alpha = 2x + 3}$ find α_{avg} of the rod.



let the rod is heated from $t_1^\circ C$ to $t_2^\circ C$,

now, $\boxed{\frac{dl}{dx} = \alpha(t_2 - t_1)}$

$$dl = (2x+3) \cdot dx \ (t_2 - t_1)$$

$$\int_0^l dl = (t_2 - t_1) \int_{x=0}^{x=l} (2x+3) \cdot dx$$

$$\Delta l = (t_2 - t_1)(l^2 + 3l)$$

$$\Delta l = l(l+3)(t_2 - t_1)$$

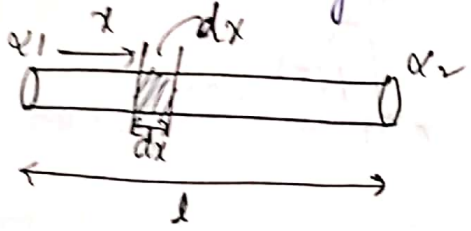
$$\frac{\Delta l}{l} = (l+3)(t_2 - t_1)$$

$$\frac{\Delta l}{l(t_2 - t_1)} = \alpha_{avg} = l + 3$$

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$\alpha_{avg} = 1+3$

4) If α-value of rod increases linearly from α_1 to α_2 along its length. Find α_{avg}.



$$\alpha_x = \alpha_1 + \left(\frac{\alpha_2-\alpha_1}{l}\right)x$$

now, $\frac{dl}{dx} = \alpha_x\,(t_2-t_1)$

$$\int_{x=0}^{x=l} dl = \int \alpha_1 + \left(\frac{\alpha_2-\alpha_1}{l}\right)x \cdot dx\,(t_2-t_1)$$

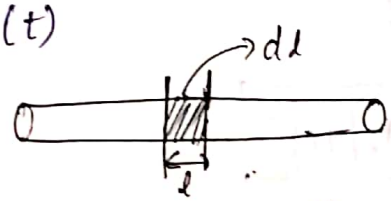
$$\Delta l = \alpha_1 l + \frac{(\alpha_2-\alpha_1)l}{2}\,(t_2-t_1)$$

$$\frac{\Delta l}{l} = \left(\alpha_1 + \frac{\alpha_2-\alpha_1}{2}\right)(t_2-t_1)$$

$$\frac{\Delta l}{l(t_2-t_1)} = \frac{\alpha_1+\alpha_2}{2} = \alpha_{avg}$$

$$\boxed{\alpha_{avg} = \frac{\alpha_1+\alpha_2}{2}}$$

(5) $\alpha = f(t)$



we have, $\frac{dl}{l} = \alpha \cdot dT$

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$$\int_{l_1}^{l_2} \frac{dl}{l} = \int_{T_1}^{T_2} f(t) \cdot dT$$

$$\ln(l_2/l_1) = \int_{T_1}^{T_2} f(t) \cdot dT$$

$$\frac{l_2}{l_1} = e^{\int_{T_1}^{T_2} f(T) \cdot dT}$$

$$\frac{l_2}{l_1} = e^{1 + \int_{T_1}^{T_2} f(T) \cdot dT + \cdots}$$

$$\frac{l_2}{l_1} = 1 + \int_{T_1}^{T_2} f(T) \cdot dT$$

$$l_2 = l_1 \left[1 + \int_{T_1}^{T_2} f(T) \cdot dT \right]$$

Ex: $\alpha = \frac{a}{T}$. Find final length of rod when heated from $T_1{}^\circ C$ to $T_2{}^\circ C$.

$$\frac{dl}{l} = \alpha \, dT$$

$$\ln(l_2/l_1) = \int_{T_1}^{T_2} \frac{a}{T} \cdot dT$$

$$\ln\left(\frac{l_2}{l_1}\right) = a \ln(T_2/T_1)$$

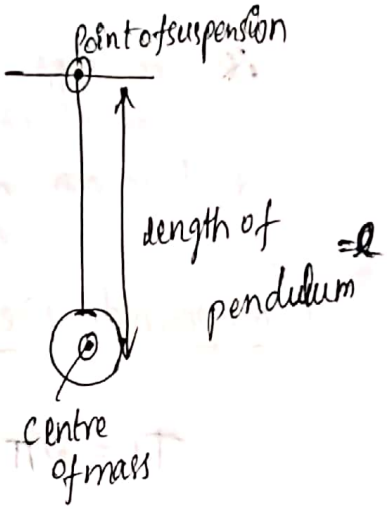
$$\frac{l_2}{l_1} = \left(\frac{T_2}{T_1}\right)^a$$

$$\boxed{l_2 = l_1 \left(T_2/T_1\right)^a}$$

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Effect of temp on Time Period of Pendulum:

① Simple Pendulum:



Time period,

$$T = 2\pi\sqrt{\frac{l}{g}}$$

* $T \propto \sqrt{l} \propto \frac{1}{\sqrt{g}}$

* As temp increases ⇒ l increases ⇒ T increases.
  ⇒ pendulum runs slowly & looses time.

* As temp decreases ⇒ l decreases ⇒ T decreases.
  ⇒ pendulum runs fast & gains time.

* we have, $T_1 = 2\pi\sqrt{\frac{l}{g}}$ (at normal temp)

& $T_2 = 2\pi\sqrt{\frac{l+\Delta l}{g}}$ (at high temp)

$$T_2 = 2\pi\sqrt{\frac{l}{g}}\left(1+\frac{\Delta l}{l}\right)^{1/2}$$

$$T_2 = T_1\left(1+\alpha(t_2-t_1)\right)^{1/2}$$

$$\boxed{T_2 = T_1\left(1+\frac{\alpha}{2}(t_2-t_1)\right)}$$

* change in oscillation per each oscillation,

$$\boxed{\Delta T = T_2 - T_1 = T_1 \times \frac{1}{2}\alpha\,(t_2-t_1)}$$

* fractional change in time period,

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$$\frac{\Delta T}{T} = \frac{T_2 - T_1}{T_1} = \frac{1}{2}\alpha(t_2 - t_1)$$

* loss (or) gain of time per second $= \frac{1}{2}\alpha(t_2 - t_1)$

* loss (or) gain of time per day $= \frac{1}{2}\alpha(t_2 - t_1) \times 86400$ sec

(2) Compound Pendulum:

$$T_1 = 2\pi\sqrt{\frac{I}{mgd}} \quad (\text{at } t_1{}^\circ C)$$

$$T_2 = 2\pi\sqrt{\frac{I + \Delta I}{mg(d+\Delta d)}} \quad (\text{at } t_2{}^\circ C)$$

$$T_2 = 2\pi\sqrt{\frac{I}{mgd}} \frac{\left(1 + \frac{\Delta I}{I}\right)^{1/2}}{\left(1 + \frac{\Delta d}{d}\right)^{1/2}}$$

$$T_2 = T_1 \frac{\left(1 + 2\alpha(t_2 - t_1)\right)^{1/2}}{\left(1 + \alpha(t_2 - t_1)\right)^{1/2}}$$

$$= T_1 \frac{\left(1 + \alpha(t_2 - t_1)\right)}{\left(1 + \alpha(t_2 - t_1)\right)^{1/2}}$$

$$= T_1 \left(1 + \alpha(t_2 - t_1)\right)\left(1 - \frac{\alpha}{2}(t_2 - t_1)\right)$$

$$= T_1 \left(1 + \alpha(t_2 - t_1) - \frac{\alpha}{2}(t_2 - t_1)\right)$$

$$T_2 = T_1\left(1 + \frac{\alpha}{2}(t_2 - t_1)\right)$$

$$\boxed{\frac{T_2 - T_1}{T_1} = \frac{\alpha}{2}(t_2 - t_1)}$$

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3 Torsional pendulum :

$$T_1 = 2\pi\sqrt{\frac{I}{C}} \quad \text{at } t_1{}^\circ C$$

$$T_2 = 2\pi\sqrt{\frac{I+\Delta I}{C}}$$

C = torsional const (doesnot change with temp)

$$T_2 = 2\pi\sqrt{\frac{I}{C}}\left(1+\frac{\Delta I}{I}\right)^{1/2}$$

$$T_2 = 2\pi\sqrt{\frac{I}{C}}\left(1+2\alpha(t_2-t_1)\right)^{1/2}$$

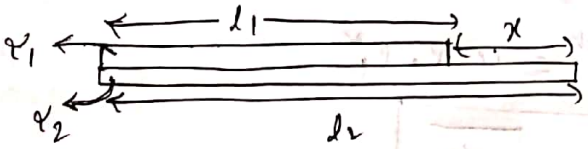
$$T_2 = T_1\left(1+\alpha(t_2-t_1)\right)$$

$$T_2 - T_1 = T_1\alpha(t_2-t_1)$$

$$\boxed{\frac{T_2-T_1}{T_1} = \alpha(t_2-t_1)}$$

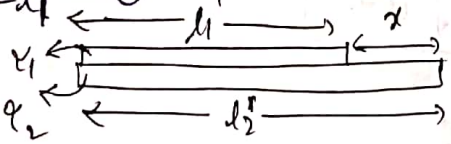
Applications :

① Difference in length of rods is same at all temp.



at $t_1{}^\circ C$

$$x = l_2 - l_1 \quad -①$$



at $t_2{}^\circ C$

$$x = l_2' - l_1' \quad -②$$

from ① & ②

$$l_2 - l_1 = l_2' - l_1'$$

$$l_2\left(1+\alpha_2(t_2-t_1)\right) - l_1\left(1+\alpha_1(t_2-t_1)\right) = l_2 - l_1$$

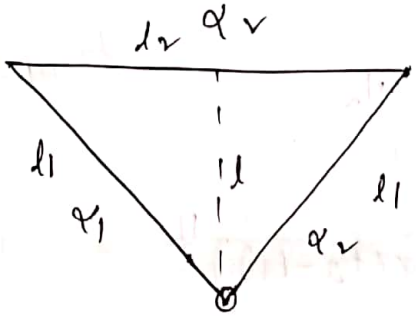
$$l_2\alpha_2(t_2-t_1) - l_1\alpha_1(t_2-t_1) = 0$$

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$l_1\alpha_1 = l_2\alpha_2$ (condn)

② Compensated pendulum :

This pendulum shows correct time at all temp.



here

$l^2 = l_1^2 - \left(\frac{l_2}{2}\right)^2$ is same at all temp.

$$\cancel{2l \cdot \frac{dl}{d\theta}}^{0} = 2l_1 \cdot \frac{dl_1}{d\theta} - \frac{2l_2}{4} \cdot \frac{dl_2}{d\theta}$$

$$\frac{l_2}{4} \cdot \frac{dl_2}{d\theta} = l_1 \cdot \frac{dl_1}{d\theta}$$

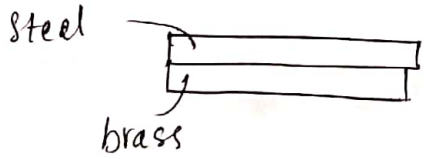
$\alpha = \frac{dl}{l\,d\theta}$

$l_2\alpha_2 l_2 = 4 l_1 \cdot \alpha_1 l_1$

$\alpha_2 l_2^2 = 4\alpha_1 l_1^2$

$$\frac{l_1}{l_2} = \sqrt{\frac{\alpha_2}{4\alpha_1}}$$

③ Grid-Iron Pendulum :



$\alpha_{st} : \alpha_B = 2:3$

$\Delta l = l\,\alpha\,\Delta\theta$

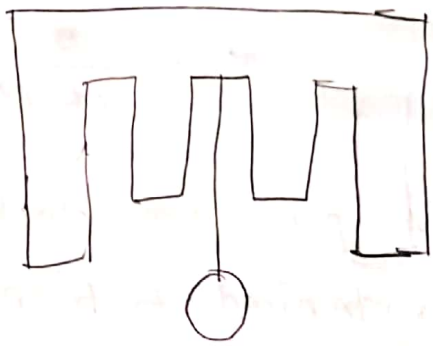
$\Delta l_{st} : \Delta l_B = 2:3$

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$\Delta l_{st} = \Delta l_B$

$l_{st} \cdot \alpha_{st} = l_B \cdot \alpha_B$

$\boxed{l_{st} : l_B = 3 : 2}$



$N_1 l_1 \alpha_1 = N_2 l_2 \alpha_2$

Ex: A pendulum clock of iron rod is connected to a small heavy bob. It is shows correct time at 20°c. How fast of slow it will go in 1 day at 40°c.

$\alpha = 12 \times 10^{-6} /^{o}c$

$T_{loss/gain} = \frac{1}{2} \alpha (t_2 - t_1) \times 86400$

$= \frac{1}{2} \times 12 \times 10^{-6} \times 10 \times 86400$

$= 12 \times 10^{-3} \times 86400$

$= 1036.8 \text{ sec}$

Ex: A clock with a metallic pendulum is 5 sec fast each day at a temp of 15°c & 10 sec slow each day at a temp of 30°c. Find the coeff of linear expansion of the metal.

let t be the temp at which pendulum shows correct reading.

$5 = \frac{1}{2} \alpha (t - 15) \times 86400 \quad -(1)$

$10 = \frac{1}{2} \alpha (30 - t) \times 86400 \quad -(2)$

$\frac{(1)}{(2)} \Rightarrow \frac{1}{2} = \frac{t - 15}{30 - t} \Rightarrow 30 - t = 2t - 30$

$60 = 3t \Rightarrow t = 20°c$

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$$\frac{5}{2} = \frac{1}{2}\alpha\,(5)\,86400$$

$$\alpha = \frac{2}{86400} = 2.314\times10^{-5}/^{\circ}C$$

Ex: A pendulum clock & a digital clock both are synchronised to beap correct time at temp 20°C in the morng on first March 2003. at 12:00 Noon temp increases to 40°C & remains const for 3 months On 1st June 2003 at 12:00 Noon temp drops to 10°C & remains const for a very long duration. Find the date time on which both the clocks will again be synchronised for a moment.

digital clock will show correct time at all temperatures.

loss of time $= \frac{1}{2}\alpha(40-20)\,86400\times92$ — ①

gain of time $= \frac{1}{2}\alpha(20-10)\times86400\times x$ — ②

$$20\times92 = 10\times x$$

$$\boxed{x = 184 \text{ days}}$$

from June 1st 2003, 184 days we get December 2nd 2003

Ex: A Iron rod of length 50cm is joined at one end to copper rod of length 100cm at 20°C. Find 1) the length of the system at 100°C. 2) α_{avg}.

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given $\alpha_{Ir} = 12 \times 10^{-6}/^{\circ}C$, $\alpha_{Cu} = 17 \times 10^{-6}/^{\circ}C$

$l_1' = l_1(1 + \alpha_1(t_2 - t_1))$

$l_2' = l_2(1 + \alpha_2(t_2 - t_1))$

1) $l_1' + l_2' = 50(1 + 12 \times 10^{-6}(80)) + 100(1 + 17 \times 10^{-6} \times 80)$

$= 50(1 + 96 \times 10^{-5}) + 100(1 + 17 \times 10^{-6} \times 80)$

$= 50 + 5 \times 96 + 10^{-4} + 100 + 17 \times 8 \times 10^{-3}$

$= 150 + (48 + 136) \times 10^{-3}$

$= 150 + 0.184$

$= 150.184$

2) $\alpha_{avg} = \frac{l_2 - l_1}{l_1(t_2 - t_1)} = \frac{0.184}{150(80)} = \frac{184}{12} \times 10^{-6}$

$\boxed{\alpha_{avg} = 15.33 \times 10^{-6}/^{\circ}C}$

Ex: A metal rod A of 50cm length expands by 0.10cm when it's temp is raised from 0°C to 100°C. Another rod B of different metal of 60cm length expands by 0.06cm for the same raise in temp. A third rod C of 50cm length is made of pieces of rods A & B placed end to end expands by 0.08cm on heating from 0°C to 50°C. Find the length of each portion of composite rod C.

$\frac{\Delta l}{l(\Delta T)} = \alpha_A$

$\frac{0.1}{50(100)} = \alpha_A \Rightarrow \alpha_A = 2 \times 10^{-5}/^{\circ}C$

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$$\frac{0.01}{100 \times 60} = \alpha_B \Rightarrow \alpha_B = 1 \times 10^{-5}/^oC$$

for c, $l_1 + l_2 = 50$

$\Delta l = \Delta l_1 + \Delta l_2$

$0.03 = l_1\alpha_1 \times 50 + l_2\alpha_2 \times 50$

$0.03 = 50\,(l_1 \times 2 + l_2) \times 10^{-4}$

$300 = (2l_1 + 0.6\, l_2)\,5$

$150 = (l_1 + 0.3\, l_2)\,5$

$150 = (50 - l_2 + 0.3\, l_2)\,5$

$0.7\, l_2 = 20$

$l_2 = \frac{200}{7}$

$l_2 = 28.7\,cm$

$l_1 + l_2 = 50$

$l_2 = 50 - l_1$

$0.03 = 50\,(2l_1 + l_2) \times 10^{-4}$

$\frac{300}{5} = 2l_1 + l_2$

$2l_1 + l_2 = 60$

$\Rightarrow l_1 = 10,\ l_2 = 40$

Ex: what should be the length of steel & copper rods. l_{st} = 10cm longer than l_{cop} at all temp. Given $\alpha_{cop} = 17 \times 10^{-6}/^oC$, $\alpha_{st} = 11 \times 10^{-6}/^oC$.

$l_1\alpha_1 = l_2\alpha_2$

$l_1 \times 17 \times 10^{-6} = (10 + l_1) \times 11 \times 10^{-6}$

$17 l_1 = 110 + 11 l_2$

$6 l_1 = 110$

$l_1 = \frac{55}{3}$

$l_1 = 18.33$

$l_2 = 28.33\,cm$

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Ex: Al whole diameter = 1m, iron sphere diameter = 1.004m

$\alpha_{Al} = 24 \times 10^{-6}/^{\circ}C$, $\alpha_{Fe} = 12 \times 10^{-6}/^{\circ}C$. Find the temp at which sphere fits in the hole. (initial temp of the system = 25°C)

$$d_{Al}(1 + \alpha_{Al}(x-25)) = d_{Fe}(1 + \alpha_{Fe}(x-25))$$

$$1 + 24 \times 10^{-6}(x-25) = 1.004 + 1.004 \times 12 \times 10^{-6}(x-25)$$

$$(24 - 1.004 \times 12) \times 10^{-6}(x-25) = 0.004$$

$$x - 25 = \frac{4000}{24 - 12.048}$$

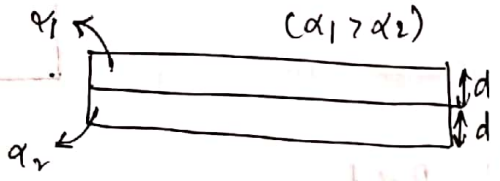
$$x - 25 = \frac{4000}{11.952}$$

$$x = 334.67 + 25$$

$$\boxed{x = 359.67}$$

Bimetallic Strips

* 2 different metals having same dimensions are revited together called bimetallic strip.
* It is used as "thermostat".

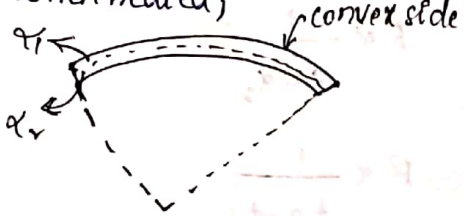


attic

when cooled,

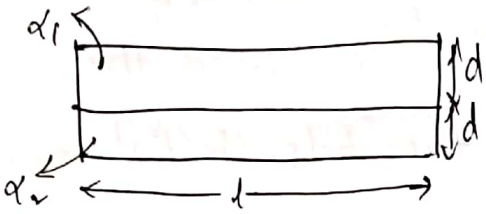


when heated,

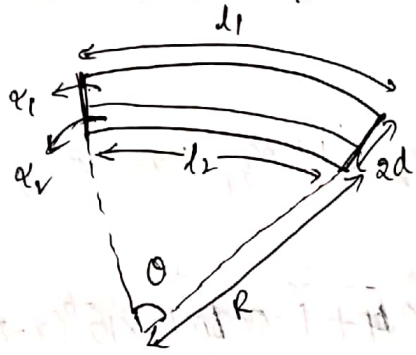


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Radius of Curvature of Bimetallic strip



when heated to $t_2°c$



From figure,

$$\theta = \frac{l_1}{R+2d} = \frac{l_2}{R}$$

$$\frac{l_1}{l_2} = \frac{R+2d}{R}$$

$$\frac{l(1+\alpha_1(t_2-t_1))}{l(1+\alpha_2(t_2-t_1))} = 1+\frac{2d}{R}$$

$$\frac{1+\alpha_1(t_2-t_1)}{1+\alpha_2(t_2-t_1)} - 1 = \frac{2d}{R}$$

$$\frac{1+\alpha_1(t_2-t_1)-1-\alpha_2(t_2-t_1)}{1+\alpha_2(t_2-t_1)} = \frac{2d}{R}$$

$$(\alpha_1-\alpha_2)(t_2-t_1) = \frac{2d}{R}$$

$$\boxed{R = \frac{2d}{(\alpha_1-\alpha_2)(t_2-t_1)}}$$

Clearly, $R \propto d$

$$R \propto \frac{1}{\alpha_1-\alpha_2}$$

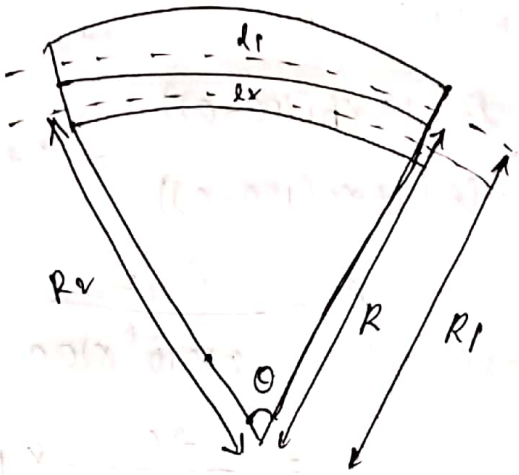
$$R \propto \frac{1}{t_2-t_1}$$

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$R_1 = R - \frac{d}{2} \Rightarrow R_1 = R + \frac{d}{2}$

$R_2 + \frac{d}{2} = R$

$\Rightarrow R_2 = R - \frac{d}{2}$

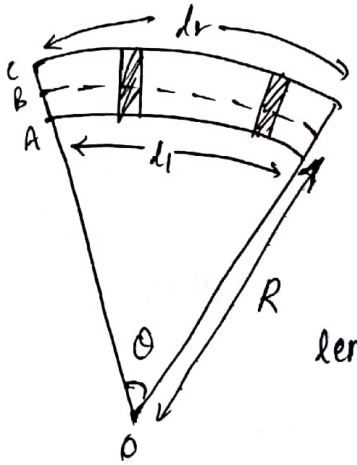


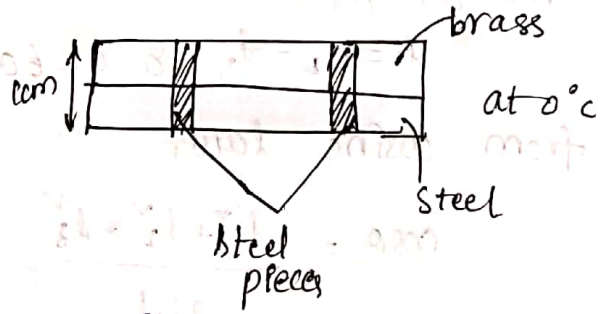
Again $\theta = \frac{l_1}{R_1} = \frac{l_2}{R_2}$

$\frac{R_1}{R_2} = \frac{l_1}{l_2}$

$\frac{R + d/2}{R - d/2} = \frac{d[1+\alpha_1(t_2 - t_1)]}{d[1+\alpha_2(t_2 - t_1)]} \Rightarrow R = \frac{d}{(\alpha_1 - \alpha_2)(t_2 - t_1)}$

Ex! Two straight thin bars one brass & one steel are joined together side by side by short steel cross pieces at 0°c, 1cm long when heated to 100°c the composite bar becomes ~~bent~~ bent into the arc of a circle. Calculate the radius of the circle $\alpha_{br} = 19 \times 10^{-6}/°C$ & $\alpha_{steel} = 11 \times 10^{-6}/°C$.





length of steel cross pieces at 100°c is

$A_C = 1(1 + \alpha_s(100 - 0))$

$= 1 + 11 \times 10^{-6} \times 100$

$= 1 + 11 \times 10^{-4}$

$\boxed{A_C = 1.0011 \text{ cm}}$

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$$\theta = \frac{l_2}{R+AC} = \frac{l_1}{R} \Rightarrow \frac{l_2}{l_1} = 1 + \frac{AC}{R}$$

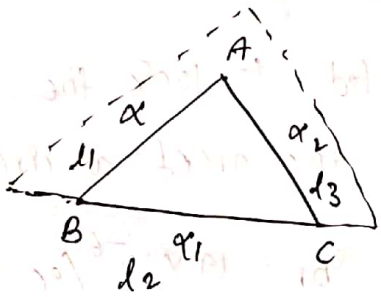
$$\frac{l_0(1+\alpha_b(100-0))}{l_0(1+\alpha_s(100-0))} = 1 + \frac{AC}{R}$$

$$R = \frac{AC/2}{8\times10^{-6}\times100}$$

$$= \frac{AC}{16}\times10^4 = \frac{1001}{16}$$

$$\boxed{R = 1252.8\,(cm)}$$

Ex: Three rods A, B, C form an equilateral $\triangle$le at 0°c. Rods AB & BC have same coeff of expansion α_1 & rod AC have α_2. If the temp of sys is increased by Δt°C, what is the change in angle θ formed by rods AB & BC.



let l_1, l_2, l_3 be the length of rods AB, BC & AC at 0°C.

$$l_1 = l_2 = l_3 \quad \& \quad \theta = 60^\circ$$

from cosine law,

$$\cos\theta = \frac{l_1^2 + l_2^2 - l_3^2}{2l_1l_2}$$

$$2l_1l_2\cos\theta = l_1^2 + l_2^2 - l_3^2$$

differentiating,

$$2l_1l_2(-\sin\theta\cdot d\theta) + 2(dl_1)l_2\cos\theta + 2l_1(dl_2)\cos\theta = 2l_1dl_1 + 2l_2dl_2 - 2l_3dl_3$$

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$$-2\ell_1\ell_2 \sin\theta \cdot d\theta + 2(\ell_1 \alpha_1 dT)\ell_2 \cos\theta + 2\ell_1(\ell_2 \alpha_1 dT)\cos\theta$$

we have, $dl = l\alpha dT$

$$= 2\ell_1\ell_1 \alpha_1 dT + 2\ell_2\ell_2 \alpha_1 dT - 2\ell_3\ell_3 \alpha_2 dT$$

$$-\frac{2\sqrt{3}}{2} \cdot d\theta + 2\alpha_1 dT \times \frac{1}{2} + 2\alpha_1 dT \times \frac{1}{2} = 2\alpha_1 dT + 2\alpha_1 dT - 2\alpha_2 dT$$

$$-\sqrt{3}\, d\theta + \alpha_1 dT + \alpha_1 dT = 2\alpha_1 dT - 2\alpha_2 dT$$

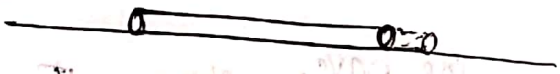
$$-\sqrt{3} d\theta = 2\alpha_1 dT - 2\alpha_2 dT$$

$$d\theta = \frac{2dT(\alpha_1 - \alpha_2)}{-\sqrt{3}}$$

$$\boxed{d\theta = \frac{2}{\sqrt{3}}(\alpha_2 - \alpha_1)dT}$$

Thermal Stress :

① when rod is placed on floor & is heated from t°c to t₂°c :-



∴ free expansion of rod, $\Delta l = l\alpha\Delta t$

∴ since there is no opposition at both ends of rod,

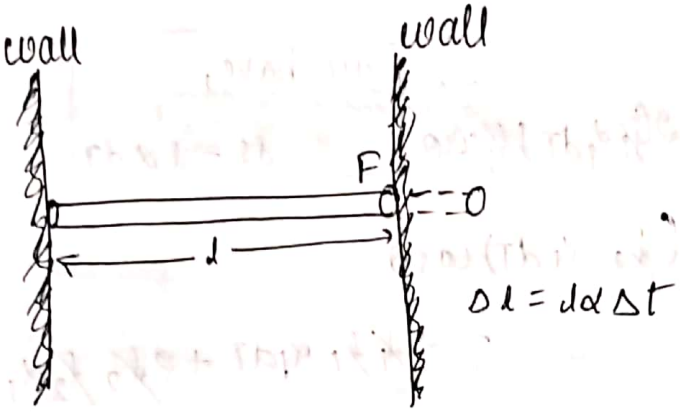
⇒ longitudinal strain = 0

⇒ thermal stress = 0

⇒ energy stored in rod = 0

⇒ displacement of C.M of body = 0.

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② 

Compressive strain in rod,

$$= \frac{l - (l + l\alpha\Delta t)}{l(1+\alpha\Delta t)}$$

$$= -\frac{\alpha\Delta t}{1+\alpha\Delta t}$$

If ϵ = modulus of elasticity (Y)

then, $$\boxed{\text{Stress} = \frac{-\epsilon\alpha\Delta t}{1+\alpha\Delta t}}$$

-ve sign indicates stress is compressive.

$$\text{Thermal stress} = \frac{-\epsilon\alpha\Delta t}{1+\alpha\Delta t \to 0}$$

$$\boxed{\text{thermal stress} = -Y\alpha\Delta t}$$

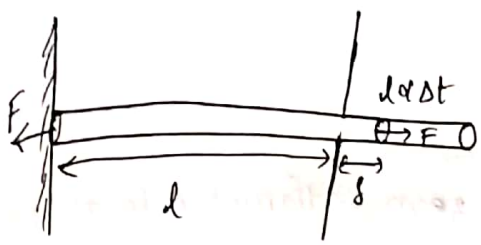
we have, $\text{stress} = \frac{F}{A}$

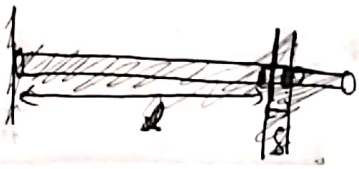
$$Y\alpha\Delta t = \frac{F}{A}$$

$$\boxed{F = YA\alpha\Delta t}$$

③ If support yields by an amount 'δ'.

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compressive strain $= \dfrac{(l+\delta) - (l + l\alpha\Delta t)}{l + l\alpha\Delta t}$

$= \dfrac{-(l\alpha\Delta t - \delta)}{l(1+\alpha\Delta t)}$

compressive stress = y × strain

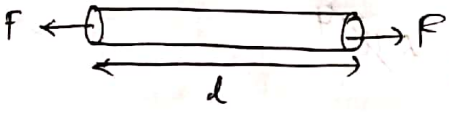
$= -y\dfrac{(l\alpha\Delta t - \delta)}{l}$

thermal stress $= y\dfrac{(l\alpha\Delta t - \delta)}{l}$

$\dfrac{F}{A} = \dfrac{y(l\alpha\Delta t - \delta)}{l}$

$$\boxed{F = \frac{yA(l\alpha\Delta t - \delta)}{l}}$$

(4)



$e = \dfrac{Fl}{Ay}$

Rod is heating and streching by force F,

Total elongation of the rod,

$\Delta l = \Delta l_1 + \Delta l_2$

$\Delta l = l\alpha\Delta t + \dfrac{F(l + l\alpha\Delta t)}{Ay}$

$= l\alpha\Delta t + \dfrac{Fl(1+\alpha\Delta t)}{Ay} \Rightarrow \boxed{\Delta l = l\alpha\Delta t + \dfrac{Fl}{Ay}}$

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$$\Delta l = l\alpha\Delta T + \frac{Fl}{Ay}$$

Ex: A steel rod of 5m long & 30cm diameter is at t_1. Find the free expansion of when temp is raised to t_2. Determine the stress & pull exerted if

1) The ends donot healed. $S = 108\ N/mm^2$, $F = 76.34\ KN$

2) The ends healed by 0.12cm. $S = 60\ N/mm^2$, $F = 42.41\ KN$

Take $y = 2 \times 10^5\ N/mm^2$ & $\alpha = 12 \times 10^{-6}/^oC$.

1)

$$S = y\alpha\Delta t$$

$$= 2 \times 10^{11} \times 12 \times 10^{-6} \times 45$$

$$= 24 \times 45 \times 10^5$$

$$= 1080 \times 10^5$$

$$= 108 \times 10^6\ N/m^2$$

$$\boxed{S = 108\ N/mm^2}$$

$$F = y\alpha\Delta t A$$

$$= 108 \times 10^6 \times 3.14 \times \frac{9}{4} \times 10^{-4}$$

$$= 27 \times 314 \times 9$$

$$= 76302\ N$$

$$\boxed{F = 76.302\ KN}$$

2)

$$S = \frac{y}{l}(l\alpha\Delta t - \delta)$$

$$= y\alpha\Delta t - \frac{y\delta}{l}$$

$$= 108 \times 10^6 - \frac{2 \times 10^{11} \times 0.12 \times 10^{-2}}{5}$$

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$= 108 \times 10^6 - 4 \times 10^6 \times 12$

$= 108 \times 10^6 - 48 \times 10^6$

$= 60 \times 10^6 \ N/m^2$

$\boxed{S = 60 \ N/mm^2}$

$F = 60 \times 10^6 \times 3.14 \times \frac{9}{4} \times 10^{-4}$

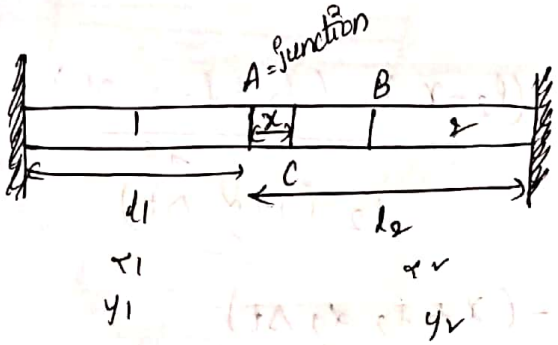
$= 225 \times 3.14$

$= 4290 \ N$

$= 42.39 \times 10^3 \ N$

$\boxed{F = 42.41 \ KN}$

*



Find the displacement of junction on heating,

for rod 1;

increase in length due to heating,

$AB = l_1 \alpha_1 \Delta t$

now due to rod 2;

compressional strain in rod 1,

$= \frac{(l_1 + x) - (l_1 + l_1 \alpha_1 \Delta t)}{l_1 (1 + \alpha_1 \Delta t)}$

$= \frac{x - l_1 \alpha_1 \Delta t}{l_1 (1 + \alpha_1 \Delta t)} = \frac{x - l_1 \alpha_1 \Delta t}{l_1}$

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$$\text{Strain}_1 = \frac{x - l_1 \alpha_1 \Delta t}{l_1}$$

$$\text{Stress}_1 = y_1 \times (\text{strain})_1$$

$$= \frac{y_1 (x - l_1 \alpha_1 \Delta t)}{l_1}$$

$$\text{Force}_1 = (\text{Stress})_1 \times A$$

$$= \frac{y_1 (x - l_1 \alpha_1 \Delta t) A}{l_1}$$

For rod 2 :

Increase in length due to heating,

$$AD = l_2 \alpha_2 \Delta t$$

due to rod 1 :

$$(\text{Strain})_2 = \frac{(l_2 - x) - (l_2 + l_2 \alpha_2 \Delta t)}{l_2 (1 + \alpha_2 \Delta t)}$$

($\alpha_2 \Delta t \to 0$)

$$(\text{Strain})_2 = \frac{-(x + l_2 \alpha_2 \Delta t)}{l_2}$$

$$(\text{Stress})_2 = \frac{-y_2 (x + l_2 \alpha_2 \Delta t)}{l_2}$$

$$(\text{Force})_2 = (\text{Stress})_2 \times A$$

$$= \frac{-y_2 A (x + l_2 \alpha_2 \Delta t)}{l_2}$$

In eqm,

$$F_1 = F_2$$

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$$\frac{Y_1 A(x - l_1\alpha_1\Delta t)}{l_1} = \frac{-Y_2 A(x + l_2\alpha_2\Delta t)}{l_2}$$

displacement of junction,

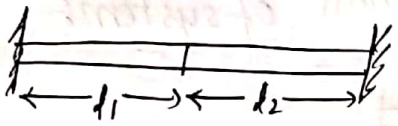
$$x = \frac{(Y_1\alpha_1 - Y_2\alpha_2)\, l_1 l_2 \Delta t}{Y_1 l_2 + Y_2 l_1}$$

points:

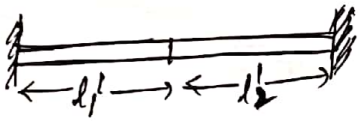
① Displacement of junction is zero

when $Y_1\alpha_1 = Y_2\alpha_2$

②



initial length of system $= l_1 + l_2$



Final length of system $= l_1' + l_2'$

clearly, $l_1 + l_2 = l_1' + l_2'$ when $l_1 = l_1'$, $l_2 = l_2'$ } when $Y_1\alpha_1 = Y_2\alpha_2$

otherwise $l_1 \neq l_1'$ & $l_2 \neq l_2'$

here, $l_1' = l_1 + l_1\alpha\Delta t - \frac{F l_1}{Y_1 A}$

& $l_2' = l_2 + l_2\alpha\Delta t - \frac{F l_2}{Y_2 A}$

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now, $\cancel{l_1' + l_2'} = \cancel{l_1 + l_2} + l_1\alpha_1\Delta t + l_2\alpha_2\Delta t - \frac{Fl_1}{AY_1} - \frac{Fl_2}{AY_2}$

$$\frac{F}{A} = \frac{l_1\alpha_1\Delta t + l_2\alpha_2\Delta t}{\frac{l_1}{Y_1} + \frac{l_2}{Y_2}}$$

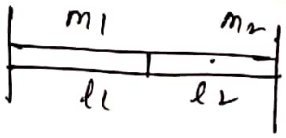
③ displacement of junction :-

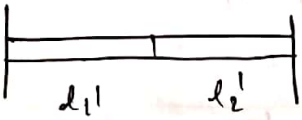
$$x = l_1' - l$$

$$= l\alpha\Delta t - \frac{F}{A}\frac{l_1}{Y_1}$$

$$x = \frac{(Y_1\alpha_1 - Y_2\alpha_2)\, l_1 l_2 \Delta t}{Y_1 l_2 + Y_2 l_1}$$

④ displacement of centre of mass of system :-





before heating,

$$(x_{cm})_i = \frac{m_1\left(\frac{l_1}{2}\right) + m_2\left(l_1 + \frac{l_2}{2}\right)}{m_1 + m_2}$$

after heating,

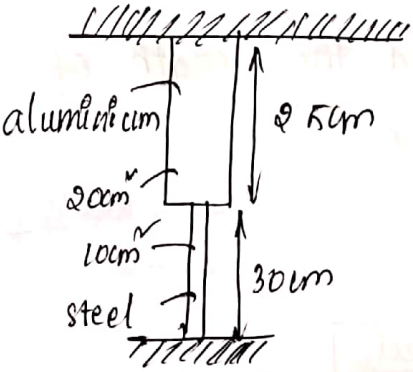
$$(x_{cm})_f = \frac{m_1\left(\frac{l_1'}{2}\right) + m_2\left(l_1' + \frac{l_2'}{2}\right)}{m_1 + m_2}$$

displacement of cm,

$$\Delta x_{cm} = (x_{cm})_f - (x_{cm})_i$$

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Ex: when the system is heated from 20°c to 70°c. Calculate stress in each rod at 70°c. take the top & bottom supports are rigid. Given $y_{Al} = 0.7 \times 10^5 N/mm^2$, $y_{st} = 2.1 \times 10^5 N/mm^2$, $\alpha_{Al} = 24 \times 10^{-6}/°c$, $\alpha_{st} = 12 \times 10^{-6}/°c$



$$l_1\alpha_1\Delta T + l_2\alpha_2\Delta T = \frac{\sigma_1 l_1}{y_1} + \frac{\sigma_2 l_2}{y_2}$$

& $F_1 = F_2$ — ①

$\sigma_1 A_1 = \sigma_2 A_2$ — ②

$$25\times10^{-2}\times24\times10^{-6}\times50 + 30\times10^{-2}\times12\times10^{-6}\times50 = \frac{\sigma_1\times25\times10^{-4}}{0.7\times10^{11}} + \frac{\sigma_2\times30\times10^{-4}}{2.1\times10^{11}}$$

$$0.7\times10^{11}(600\times50\times10^{-6} + 360\times50\times10^{-6}) = 25\sigma_1 + 10\sigma_2$$

$$7(30000 + 18000)\times10^4 = 25\sigma_1 + 10\sigma_2$$

$$7\times48\times10^7 = 25\sigma_1 + 10\sigma_2$$

② ⇒

$$\sigma_1\times2 = \sigma_2\times1 \rightarrow$$

$$\boxed{2\sigma_1 = \sigma_2} \qquad 10\sigma_2 = 20\sigma_1$$

$$\frac{7\times48}{45}\times10^7 = 45\sigma_1$$

$$\sigma_1 = \frac{49\times16}{15}\times10^7$$

$$\sigma_1 = \frac{98}{15}\times10^7\ N/m^2$$

$$\sigma_1 = \frac{7\times48}{45}\times10^7\ N/m^2$$

$$= \frac{7\times48\times10^2}{45}\ N/mm^2$$

$$\sigma_1 = \frac{14\times48}{9}$$

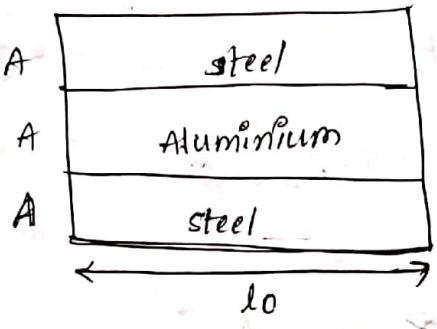
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$$\sigma_1 = \frac{14 \times 48}{9} = 74.68 \text{ N/mm}^2$$

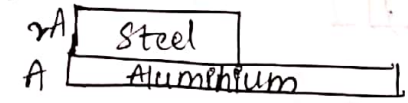
$$\sigma_2 = \frac{2 \times 14 \times 48}{9} = 149.36 \text{ N/mm}^2.$$

Ex: All rods are in a state of 'o' tension at $0^\circ c$

Find the length of the system when temp is raised to θ.



Sol:



here,

$$\Delta l_s = l_s \alpha_s \theta + \frac{F l_0}{2 A y_s} \quad \text{— (1)}$$

$$\Delta l_a = l_a \alpha_a \theta - \frac{F l_0}{A y_a} \quad \text{— (2)}$$

we have,

$$\Delta l_s = \Delta l_a$$

$$\Rightarrow l_0 \alpha_s \theta + \frac{F l_0}{2 A y_s} = l_0 \alpha_a \theta - \frac{F l_0}{A y_a}$$

$$\boxed{\frac{F}{A} = \frac{\theta(\alpha_a + \alpha_s)}{\frac{1}{2y_s} + \frac{1}{y_a}}}$$

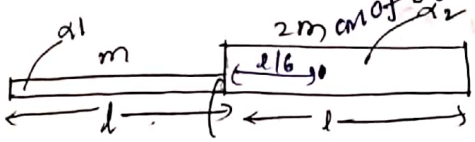
Finally,

length of system at $0^\circ c$

$$l = l_0 + \Delta l_s$$

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$$l = l_0\left[1 + \frac{(\gamma_a \alpha_a + 2\alpha_s \gamma_s)\theta}{\gamma_a + 2\gamma_s}\right]$$

Ex: 

Find displacement of junction on heating from $\theta_1°C$ to $\theta_2°C$.

Here C.M of system remains same on heating.

position of C.M of system from left end,

$$x_{cm} = \frac{m\left(\frac{l}{2}\right) + m\left(l + \frac{l}{2}\right)}{m + 2m}$$

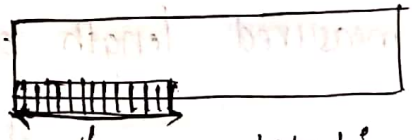
$$= \frac{\frac{ml}{2} + 3ml}{3m}$$

$$= \frac{7l}{6} = l + \frac{l}{6}$$

displacement of junction $= \frac{l}{6}\alpha_2 \Delta\theta$ to left.

Metal Scales & tapes :

① metal scale gives correct reading at its "calibration temp" (C.T)



*when metal scale is at a temp $t_2°C$ ($> t_1°C$).

* metal scale expands and measured length is less than actual length or true length.

* If l is measured length at $t_2°C$, then error in measurement, $\boxed{\text{Error} = l\alpha\Delta t = \Delta l}$

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$\therefore$ actual length $= l + l\alpha\Delta t$

$$\boxed{l_{TV} = l(1+\alpha\Delta t)}$$

* If l is measured at $t_2°c (< t_1°c)$

$\Rightarrow$ metal scale contracts.

measured length is greater than actual length.

$\therefore$ actual length $= l - l\alpha\Delta t$

$$\boxed{l_{TV} = l(1-\alpha\Delta t)}$$

(2) If both metal scale & rod expands on heating.

let $t_1°c$ = calibrated temp of scale

measured length of rod at $t_1°c$ = true length = L_{TV}

now the system (both scale & rod) is heated to $t_2°c$

* length of rod at $t_2°c$

$$= l_{TV}\left(1 + \alpha_r(t_2 - t_1)\right)$$

* length of 1cm length of scale at $t_2°c$

$$= 1\times\left(1 + \alpha_{scale}(t_2 - t_1)\right)$$

* measured length of rod at $t_2°c$

$$l_{measured} = \frac{l_{TV}\left(1 + \alpha_r(t_2 - t_1)\right)}{1\times\left(1 + \alpha_{scale}(t_2 - t_1)\right)}$$

$$= l_{TV}\left(1 + \alpha_{rod}(t_2 - t_1)\right)\left(1 + \alpha_{scale}(t_2 - t_1)\right)^{-1}$$

$$= l_{TV}\left(1 + \alpha_{rod}(t_2 - t_1)\right)\left(1 - \alpha_{scale}(t_2 - t_1)\right)$$

$$\boxed{l_{measured} = l_{TV}\left(1 + (\alpha_{rod} - \alpha_{scale})(t_2 - t_1)\right)}$$

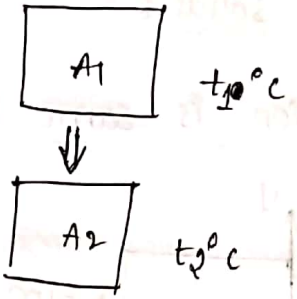
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(or) $\boxed{l_{TV} = l_{measured}\left(1-(\alpha_{rod}-\alpha_{scale})(t_2-t_1)\right)}$

Areal Expansion

└ thin sheets, plates, discs.

* we have,



$$\frac{dl}{l} \propto dT$$

$$\frac{dA}{A} = \beta\, dT$$

└ coeff of areal expansion

$$\int_{A1}^{A_V} \frac{dA}{A} = \beta \int_{T1}^{T_V} dT$$

$$\ln\left(\frac{A_V}{A_1}\right) = \beta(T_2-T_1)$$

$$\frac{A_2}{A_1} = e^{\beta(T_2-T_1)}$$

$$\boxed{A_2 = A_1(1+\beta(t_2-t_1))} \Rightarrow \boxed{\beta = \frac{A_2-A_1}{A_1(t_2-t_1)}\ /^\circ C}$$

$A_2 - A_1 = A_1\beta(t_2-t_1)$ — change in area

$\frac{\Delta A}{A_1} = \beta(t_2-t_1)$ — fractional change in area

$\frac{\Delta A}{A_1} \times 100 = \beta(t_2-t_1)\times 100$ — % of fraction change in area.

Volume Expansion

└ (or) cubical expansion

└ cubes, spheres, cylinder

* we have,

$V_2 = V_1(1+\gamma(t_2-t_1))$ — coeff of volume expansion.

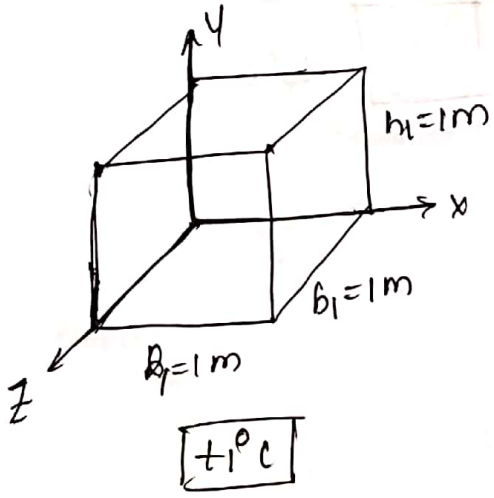
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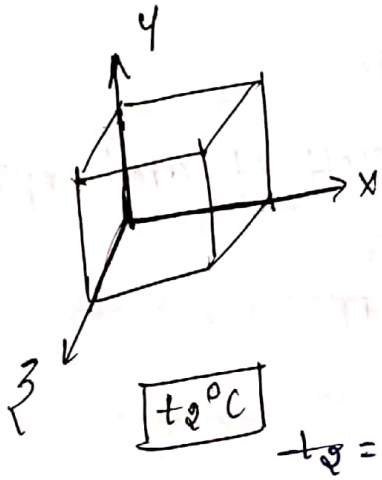
$$\frac{V_2 - V_1}{V_1} = \gamma(t_2 - t_1)$$

$$\& \quad \frac{V_2 - V_1}{V_1} \times 100 = \gamma(t_2 - t_1) \times 100$$

Isotropic solids :

Expansion is same in all directions.





$t_2 = t_1° + 1°C$

initial area, $A_1 = l_1 b_1 = 1 \times 1 = m^2$

Initial volume, $V_1 = l_1 b_1 h_1$

$= 1 \times 1 \times 1 = 1m^3$

* final length,

$$l_2 = 1 + \alpha(t_2 - t_1)$$

$$= 1 + \alpha$$

$$b_2 = 1 + \alpha$$

$$h_2 = 1 + \alpha$$

* Final area,

$$A_2 = l_2 b_2 = (1+\alpha)^2 = 1 + 2\alpha$$

* Final volume,

$$V_2 = l_2 b_2 h_2 = (1+\alpha)^3 = 1 + 3\alpha$$

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now, $\beta = \frac{A_2 - A_1}{A_1(t_2 - t_1)}$

$$\beta = \frac{1 + 2\alpha - 1}{1 \times 1}$$

$$\boxed{\beta = 2\alpha}$$

& $\gamma = \frac{V_2 - V_1}{V_1(t_2 - t_1)} = \frac{1 + 3\alpha - 1}{1 \times 1}$

$$\boxed{\gamma = 3\alpha}$$

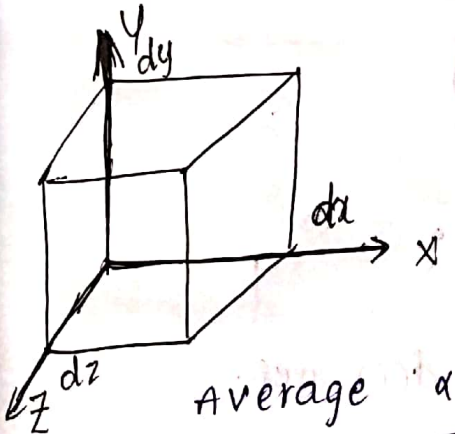
Relation b/w α, β, γ:

$$\boxed{\beta = 2\alpha} \quad , \quad \boxed{\gamma = 3\alpha}$$

$$\boxed{\alpha : \beta : \gamma = 1 : 2 : 3}$$

Anisotropic solids:

Expansion is not same in all directions.



Average α-values

$$\boxed{\alpha = \frac{\alpha_x + \alpha_y + \alpha_z}{3}}$$

$$3\alpha = \alpha_x + \alpha_y + \alpha_z$$

$$\boxed{\gamma = \alpha_x + \alpha_y + \alpha_z}$$

now,

$$\beta_{xy} = \alpha_x + \alpha_y \quad , \quad \beta_{yz} = \beta_y + \beta_z \quad , \quad \beta_{zx} = \beta_z + \beta_x$$

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Average, $\beta = \frac{\beta_{xy} + \beta_{yz} + \beta_{zx}}{3}$

$$\beta = \frac{2}{3}(\alpha_x + \alpha_y + \alpha_z)$$

$$\boxed{\beta = \frac{2}{3}\gamma}$$

Variation of density of solid with temperature :-

mass of solid,

$$m = vd$$

mass remains same on heating.

$$m = v_1 d_1 \text{ at } t_1^\circ C$$

& $m = v_2 d_2$ at $t_2^\circ C$

$$v_1 d_1 = v_2 d_2$$

$$v_1 d_1 = v_1(1 + \gamma(t_2 - t_1))\, d_2$$

$$d_1 = d_2 (1 + \gamma(t_2 - t_1))$$

$$\boxed{d_2 = \frac{d_1}{1 + \gamma(t_2 - t_1)}}$$

as temp increases $\Rightarrow$ density decreases.

* when a solid sphere & hollow sphere having same mass & same material are heated while supplied equal quantities of heat.



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here, $dQ = msdT$

here dQ, m, s are const.

$\Rightarrow dT$ = same for both the spheres.

we have,

$$\Delta R = R\alpha\Delta T$$

R is more for hollow sphere.

ΔR is more for hollow sphere.

i.e, hollow sphere expands more than solid sphere.

and $\boxed{\frac{\Delta R}{R} = \alpha\Delta T}$ same for hollow sphere

i.e, $\frac{\Delta V}{V} =$

* when a solid sphere & hollow sphere having same size (radius) & same material are heated by equal amounts of heat.



we have, $dQ = msdT$

$$dT = \frac{dQ}{ms}$$

since mass of solid sphere is more than mass of hollow sphere then dT is more for hollow sphere.

we have, $\Delta R = R\alpha\Delta T$

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$dR \propto dT$

i.e, expansion is more for hollow sphere than for solid sphere.

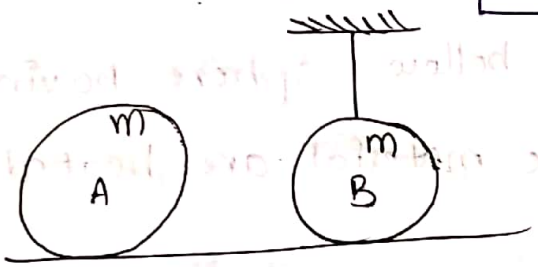
$\frac{dR}{R} = \alpha dT$ is also more for hollow sphere.

* solid sphere & hollow sphere have same dimensions & same raise in temp then they expands equally.

* Thin rod & thick rod have same mass & same raise in temp then thin rod expands more.

because length of thin rod is more, so change in will be more (from $\Delta l = l \alpha \Delta t$)

Ex!



A & B are placed on insulating plate. B is hanged & A is in rest on plate.

Then in which body more temp is raised.

* In A :

when it is heated, A expands which causes change in position of centre of mass towards up.

after heated: A 

⇒ P.E is increased with raise in temperature.

* In B :

when it is heated, B expands & moves docon.

after heated: B 

⇒ P.E is decreased with raise in temp.

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